

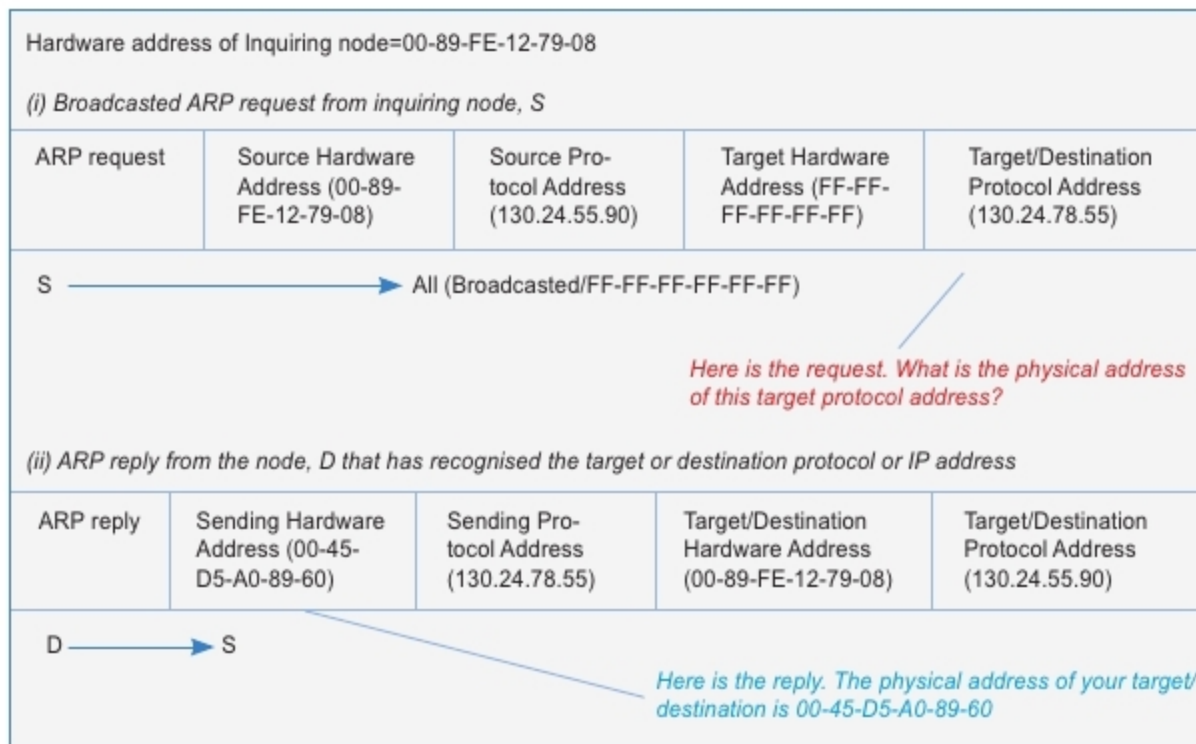


Figure 3: ARP reply and request

that is in communication with the same process at the sender.

The 16-bit length field refers to the length of the whole UDP pack in bytes. Here, header fields are fixed, unlike in TCP. Data field is then determined from length field minus header field.

The 16-bit checksum field is used to check errors in the frame. Checksum measure is taken over the entire frame. When the receiver gets UDP frame, if calculated checksum is different from that of the transmitter, the receiver will detect the error.

A few commonly-used ports for UDP applications are 53 for DNS (domain name server), 123 for NTP (network time protocol), 161 for SNMP (simple network management protocol) and 2049 for NFS (network file system).

Above TCP/IP or UDP/IP suite is the application/higher layer. It may be layer 4 if the suite is taken as one single layer, or layer 5 if the suite is taken as two separate layers. From the Internet point of view, the suite is of single layer. Thus, layer 4 is the application layer, and this corresponds to the remaining higher layers of OSI/ISO protocol. Services of the layer include file transfer protocol (FTP), remote login, computer mail and program-to-program communication using socket programming interface and remote procedure call (RPC).

FTP allows a user, computer or terminal to get (or send) files to and from another computer. It is a tool of the Internet. Remote login is based on TELNET (network terminal protocol), which is another tool of the Internet. It allows any user to log into any computer in the network. Computer mail allows a user to send mails to another computer in the network—one classic example of this is e-mail. Each of the layers, while performing their functions, adds a header to the message/datagram at transmitting side—these headers are removed by the corresponding layer at receiving side. Layer-to-layer function is a peer process.

TCP/IP is an internetworking protocol. The Internet is an interconnected network of wide varieties of networks. Interconnection among different networks (often incompatible) is done by several methods, as given below.

- By repeaters that cover only the physical layer. An example would be an Ethernet to Ethernet connection.
- By bridges that cover physical and data link layers, corresponding to OSI/ISO protocol. An example would be connecting Ethernet LAN to token bus LAN.
- By routers that cover physical, data

link and network layers (complete node coverage/node in OSI/ISO reference, which equals physical + data link + network layer). An example would be any IEEE802 series LAN to any X.25 WAN or MAN connection.

- By gateways that are used for connection networks of different protocols. A gateway covers all layers of a protocol. These are logically full protocol converters. An example would be a connection between an OSI/ISO network and a DNA network.

Repeaters, bridges and routers are used when interconnecting networks are of the same protocol. Gateways, repeaters, bridges and routers are all made of computing resources. In fact, these are all computers. Complexity of systems increases as one moves from repeaters to gateways. Bridges and routers can be defined as simple gateways. All these devices interconnect incompatible networks physically. The Internet does so logically for all networks globally.

Other Internet protocols

Address resolution protocol (ARP):

IP addresses are unique logical addresses, each assigned to a physical machine, system or node. An IP layer releases IP packets with IP logical addresses to the physical layer for the purpose of transport. The physical layer treats each IP packet as a data pack and encapsulates the pack into the physically-enabled transport packet commensurate to the physical network. For example, if the physical transport network is Ethernet LAN, IP packet as a data pack is encapsulated in the Ethernet frame. In making of the Ethernet frame, destination address required in the frame must be the physical address of the destination node.

Physical address is the address provided by network interface card (NIC) of the node. This is the only address that the physical transport network can recognise for transporting